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FA24-BCS-113

CSC336 Web Technologies — Section B Lab Midterm

Book Corner: category navigation, checkout, and cart

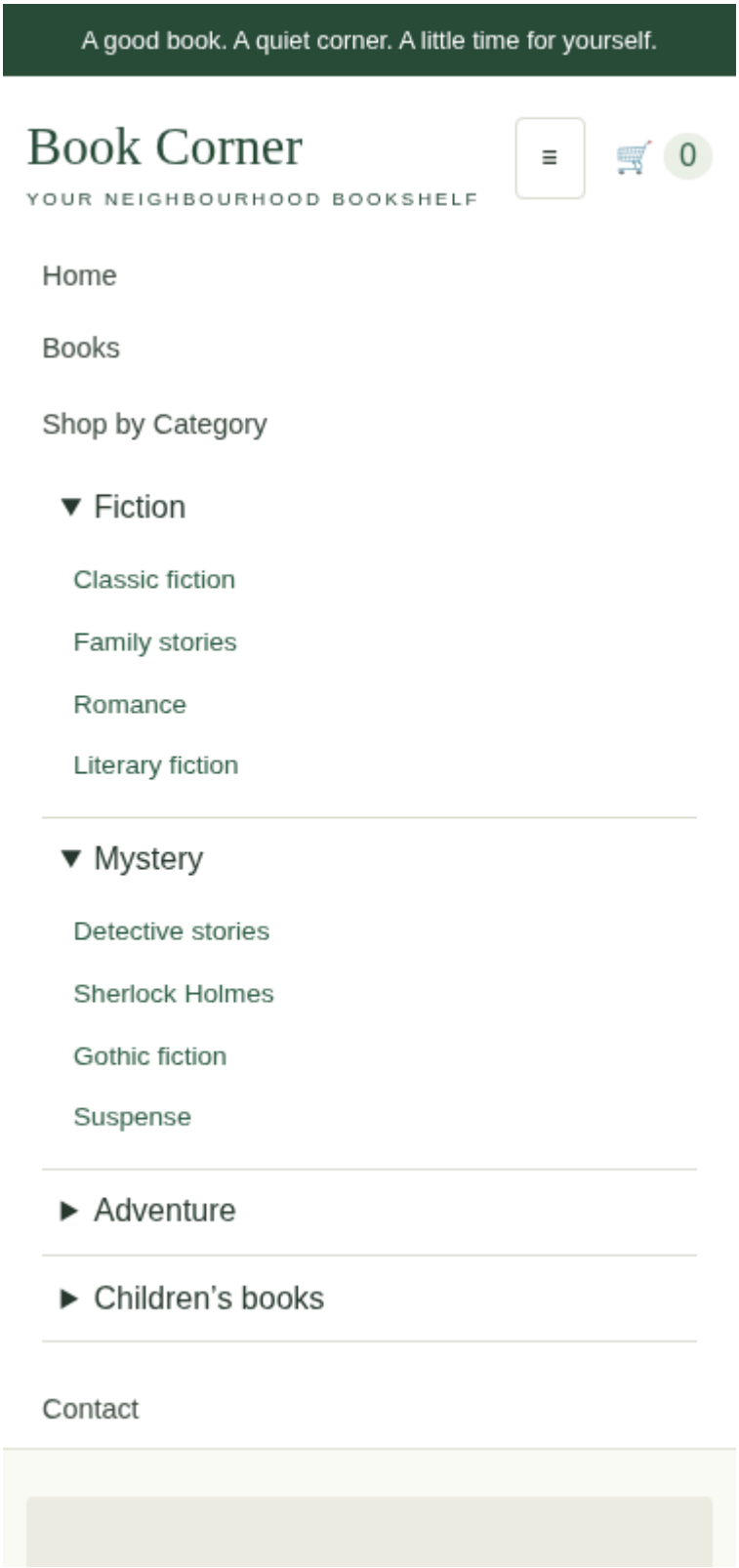
Exam date: 29 October 2026

N = 13 · BP = 613px · H = 91° · K = 4 · Cart limit = 3 copies per product

The screenshots come first, followed by selectable HTML, CSS, and JavaScript under a heading for each task.

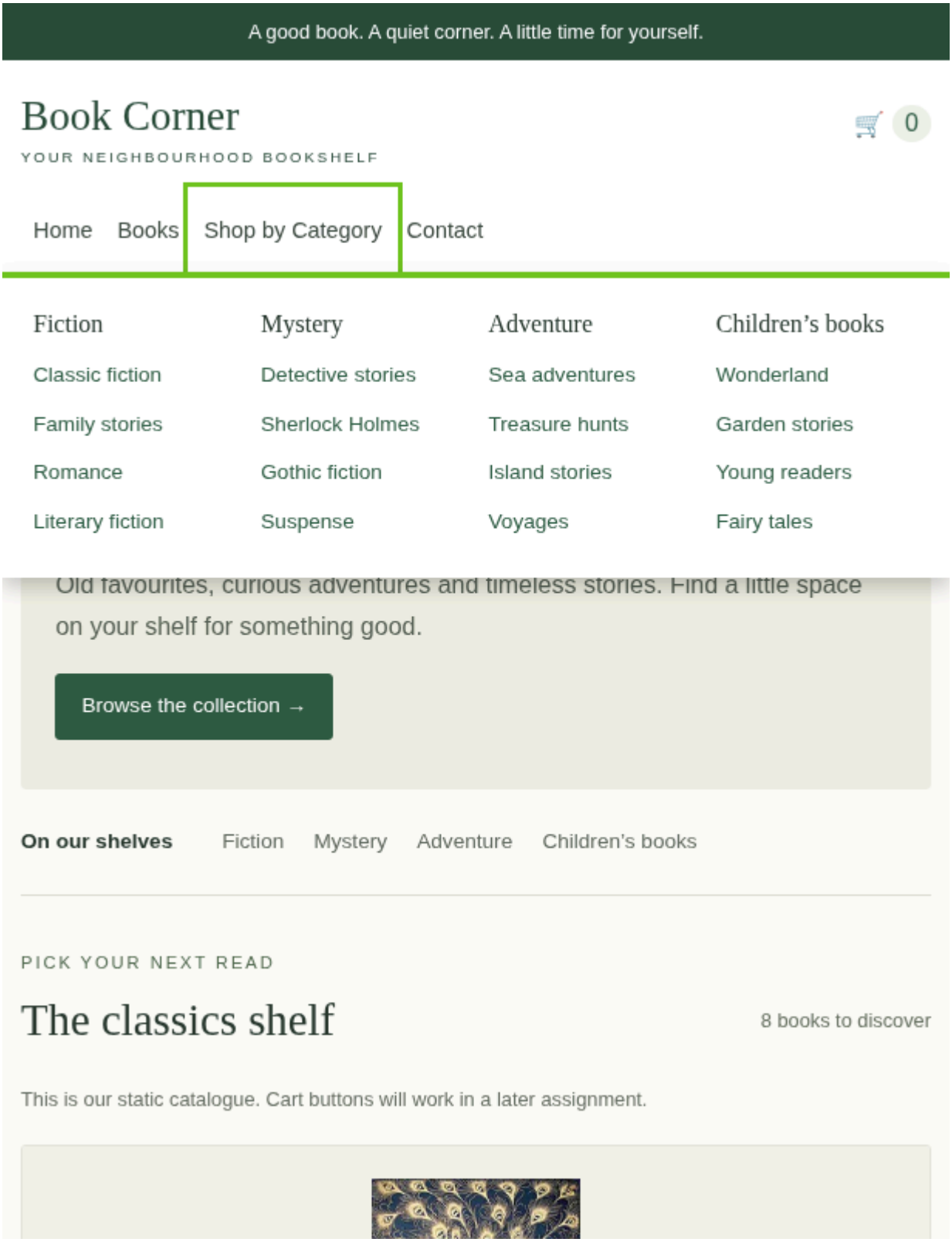
Task 1 — Category menu at 375px

Two independent mobile accordions open.



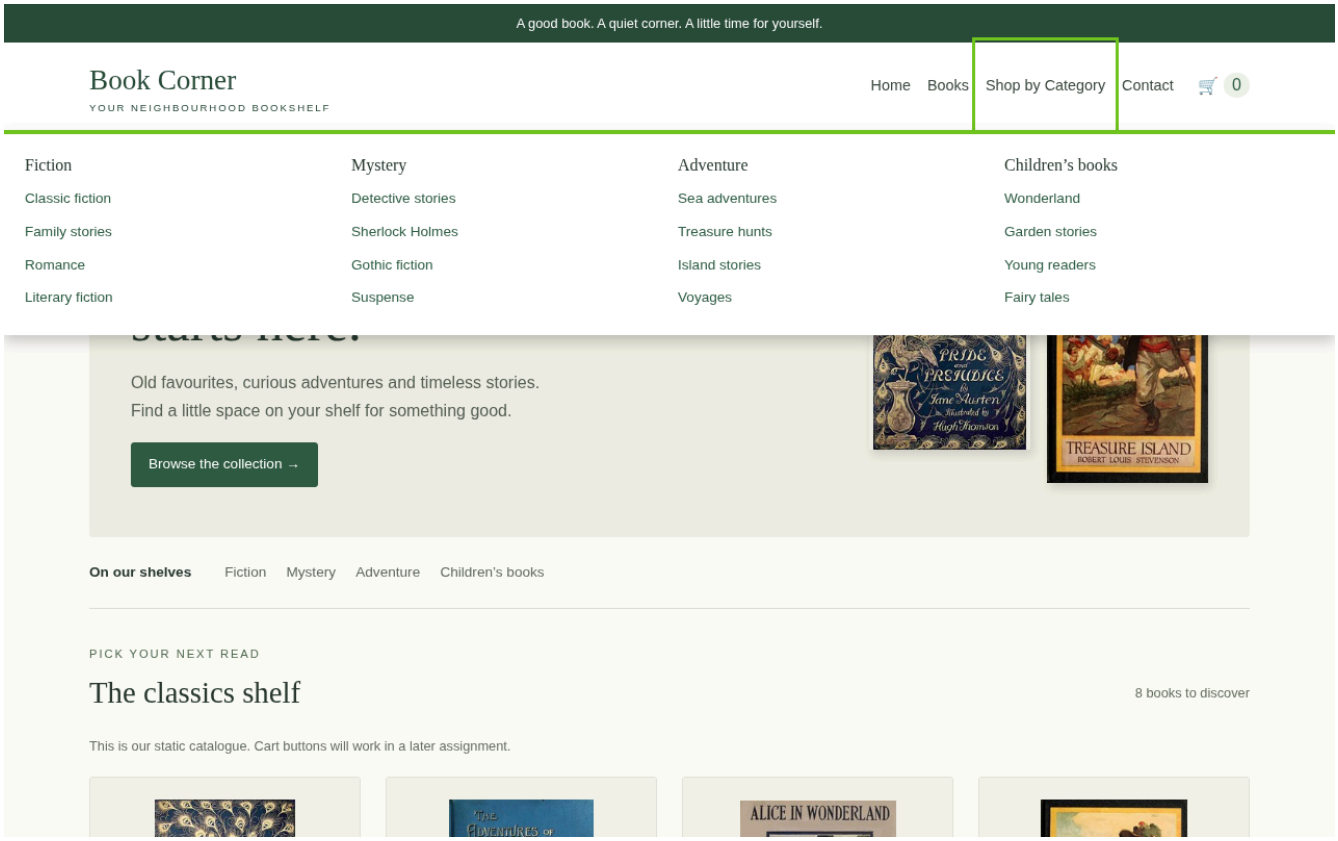
Task 1 — Category menu at 613px

Four-column mega-menu open using keyboard focus; content is overlaid.



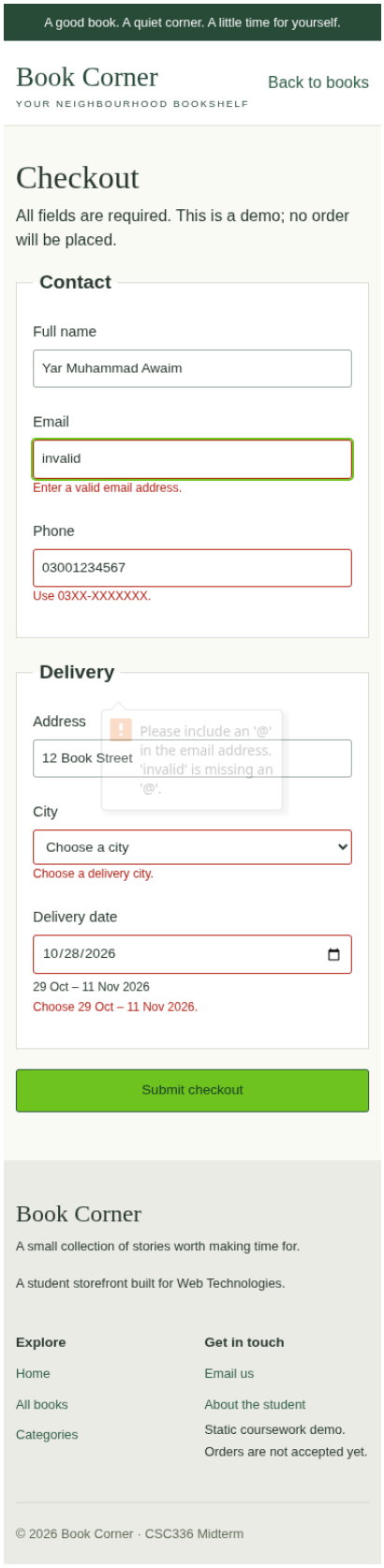
Task 1 — Category menu at 1280px

Four-column mega-menu open using keyboard focus; content is overlaid.



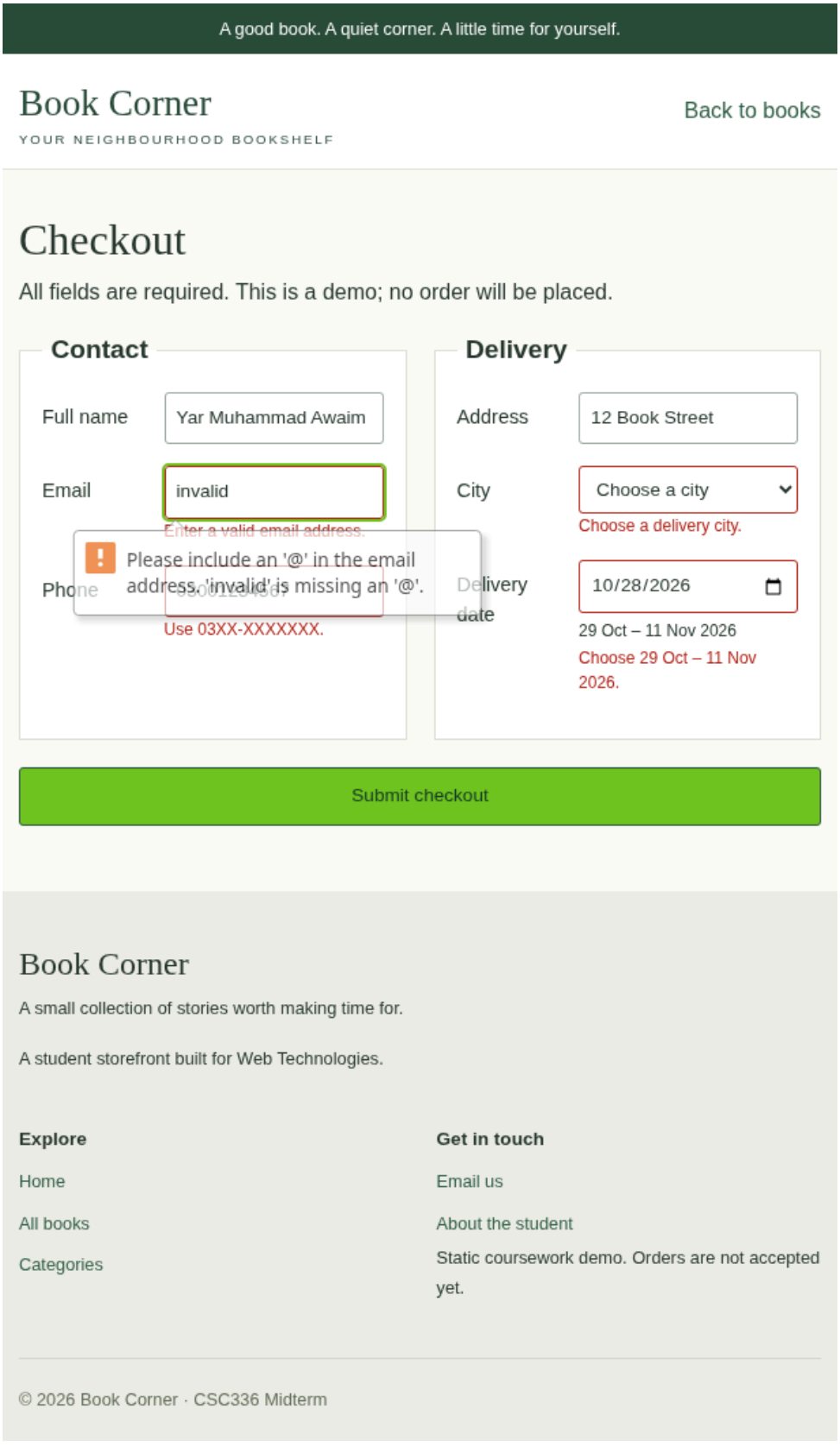
Task 2 — Checkout at 375px

HTML validation blocks invalid values. Red borders and messages appear after submission was attempted.



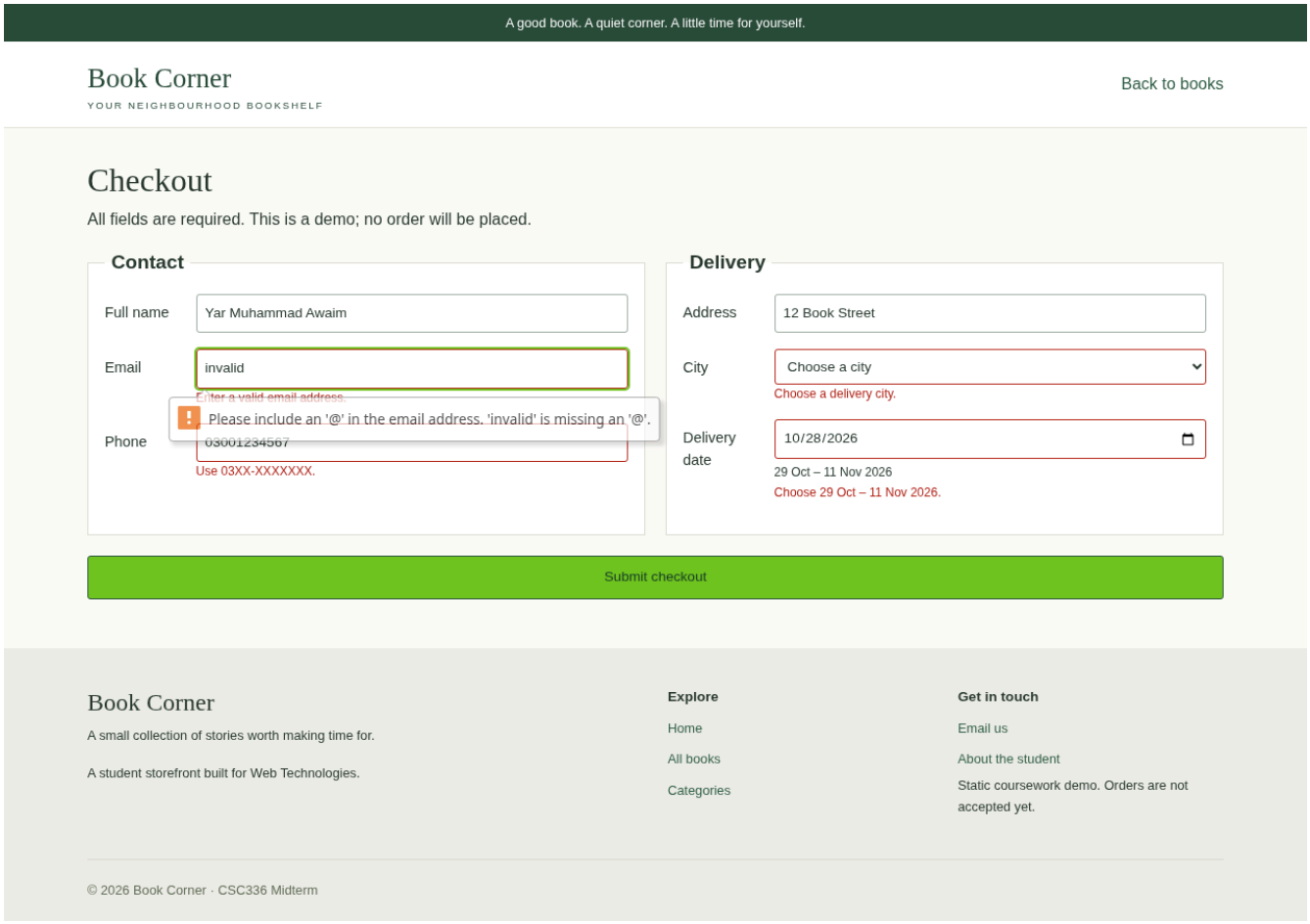
Task 2 — Checkout at 613px

HTML validation blocks invalid values. Red borders and messages appear after submission was attempted.



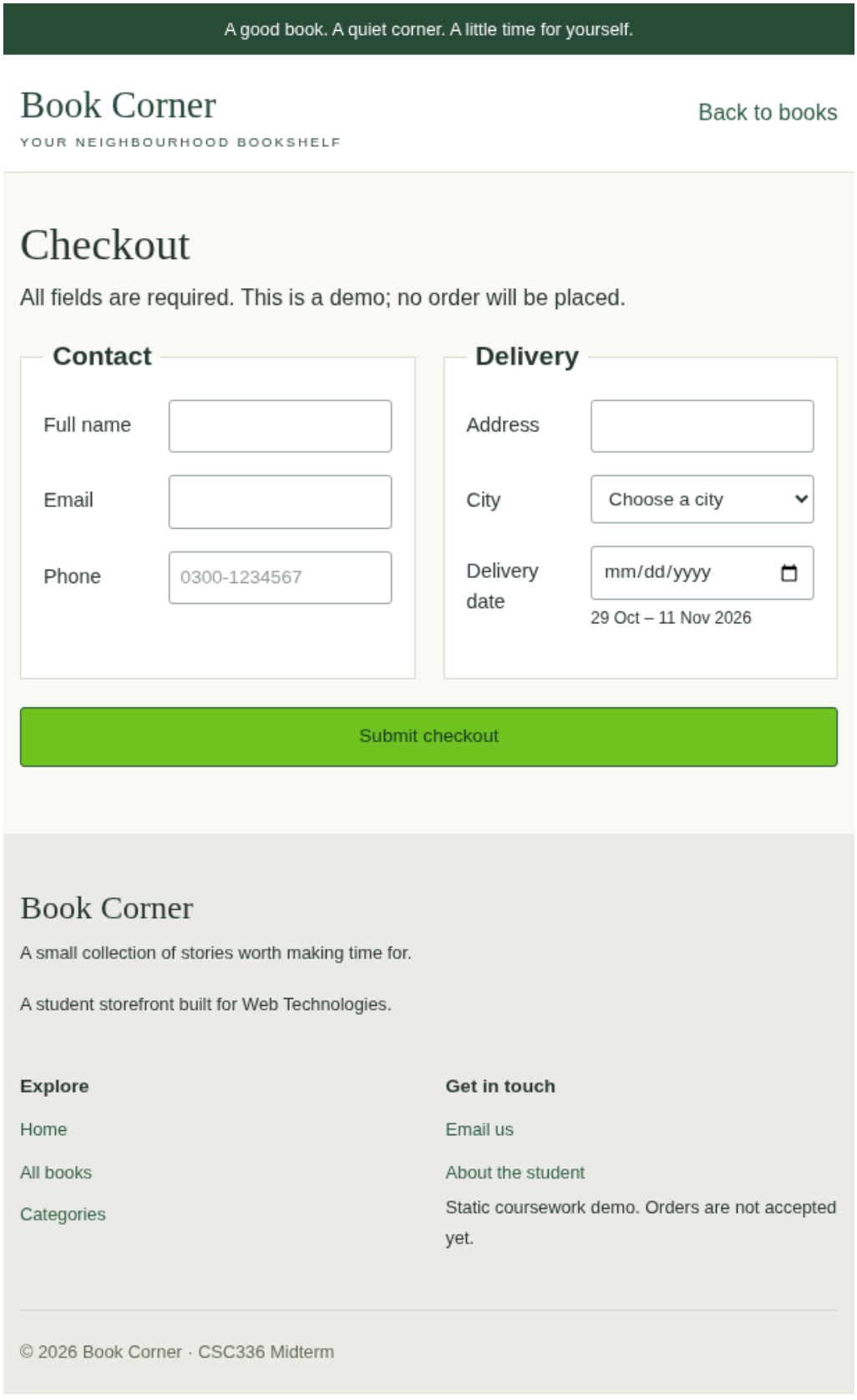
Task 2 — Checkout at 1280px

HTML validation blocks invalid values. Red borders and messages appear after submission was attempted.



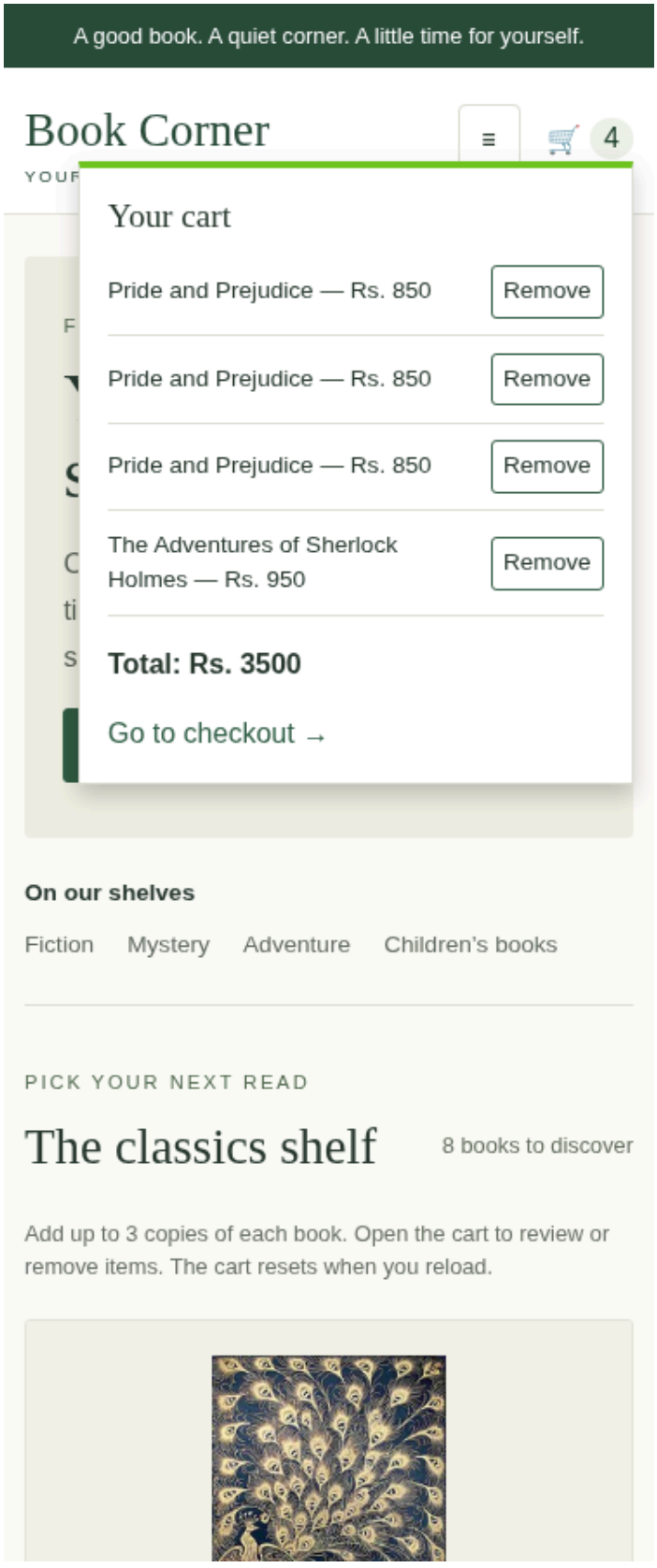
Task 2 — Fresh checkout at 613px

No error messages on load; fieldsets side by side and labels to the left.



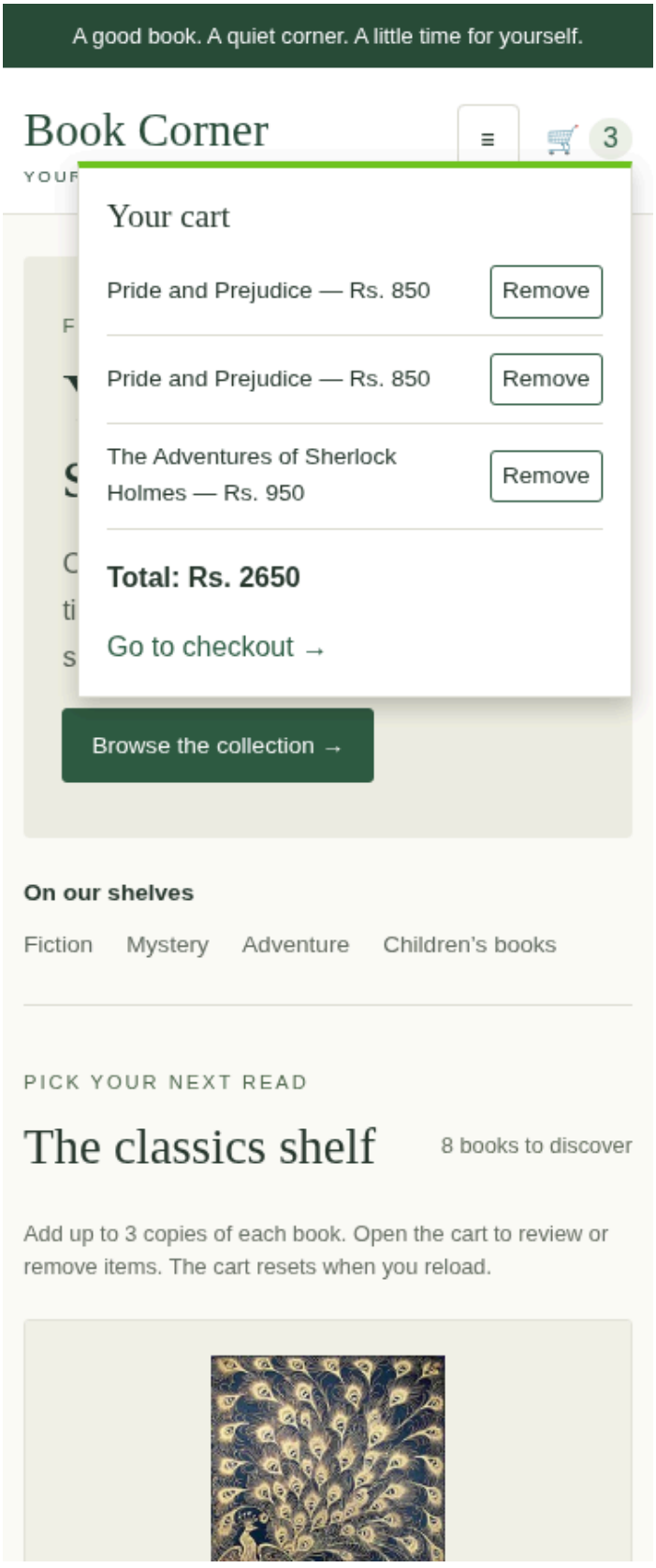
Task 3 — Cart after additions at 375px

Three Pride and Prejudice copies and one Sherlock Holmes copy: 4 items, total Rs. 3500.



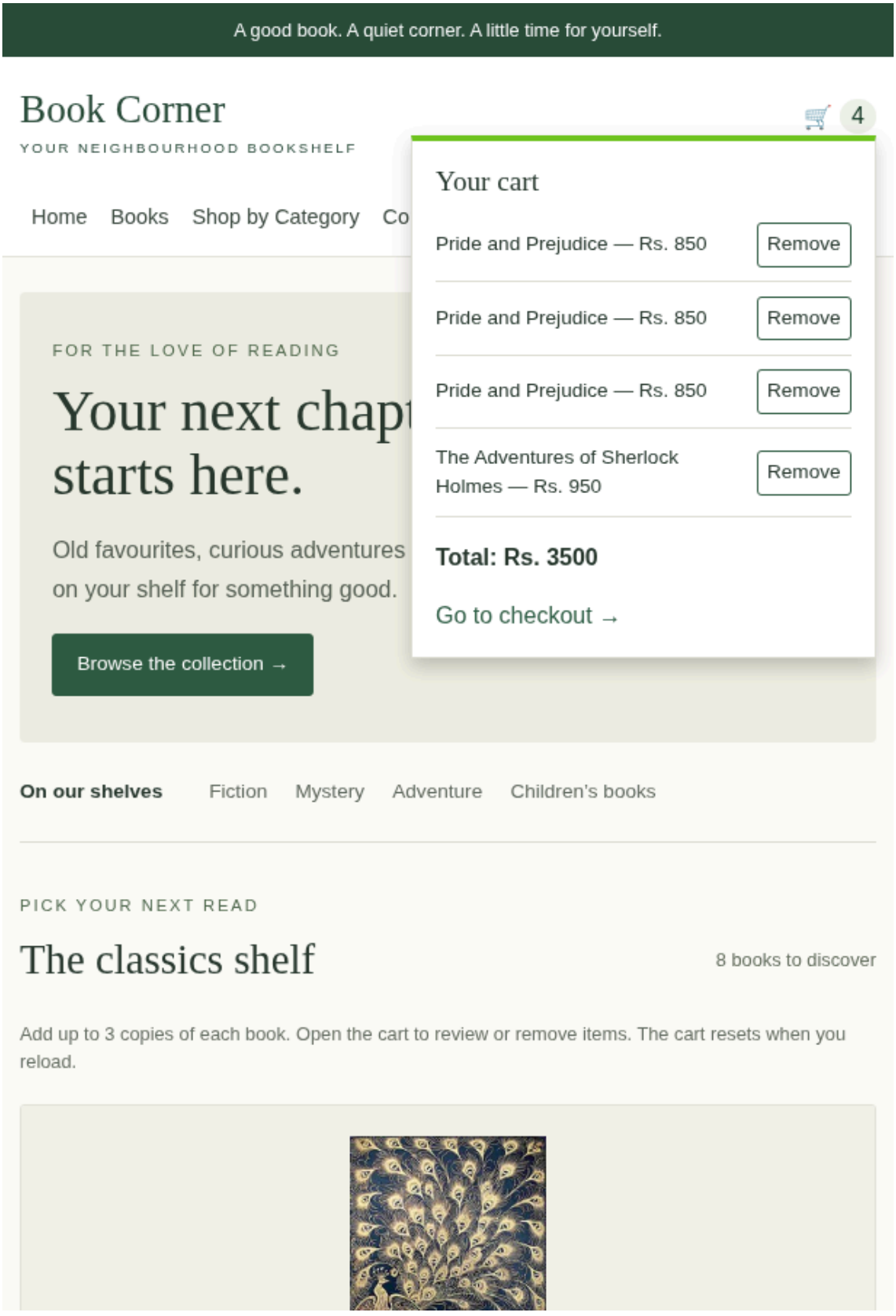
Task 3 — Cart after removal at 375px

One Pride and Prejudice copy removed: 3 items, total Rs. 2650. That book can be added again.



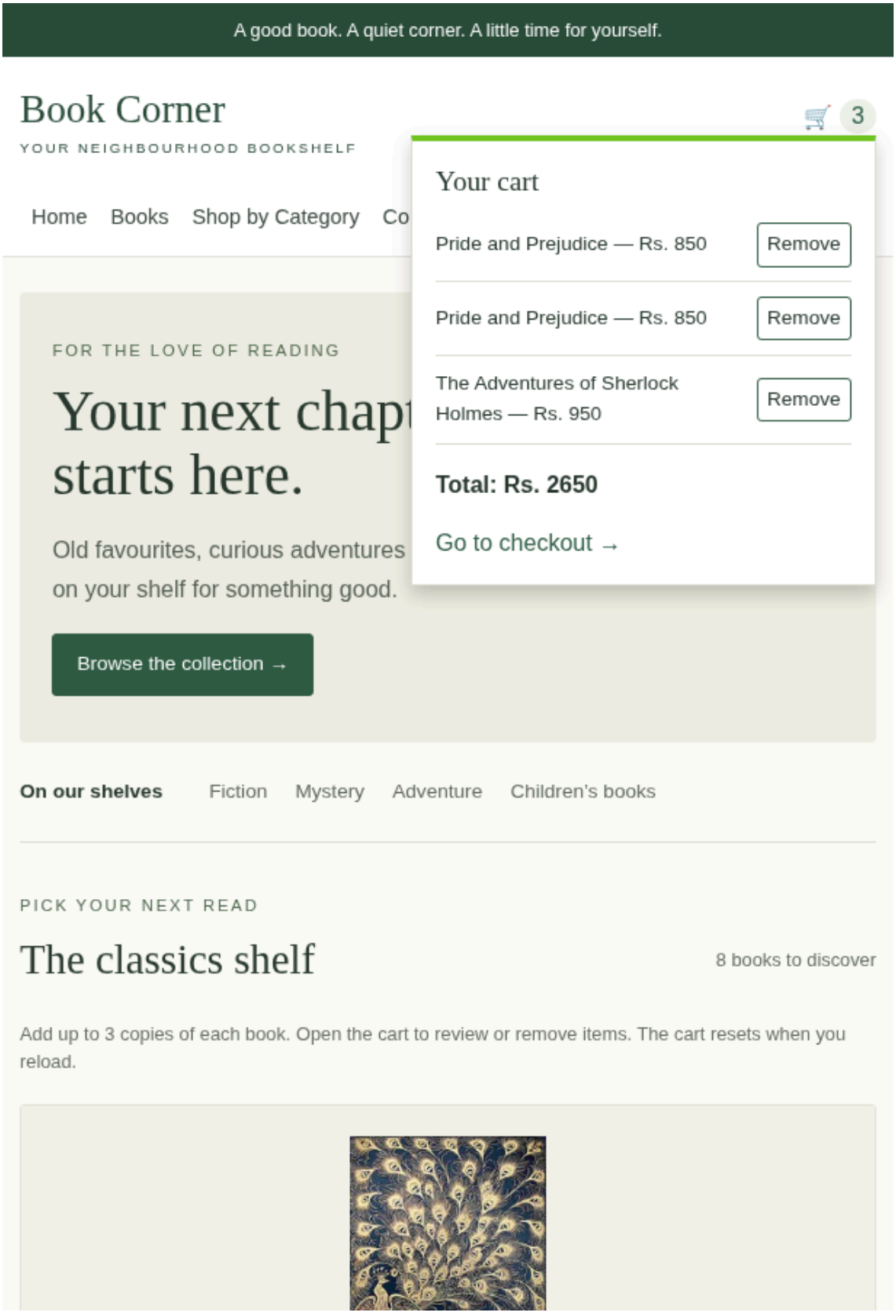
Task 3 — Cart after additions at 613px

Three Pride and Prejudice copies and one Sherlock Holmes copy: 4 items, total Rs. 3500.



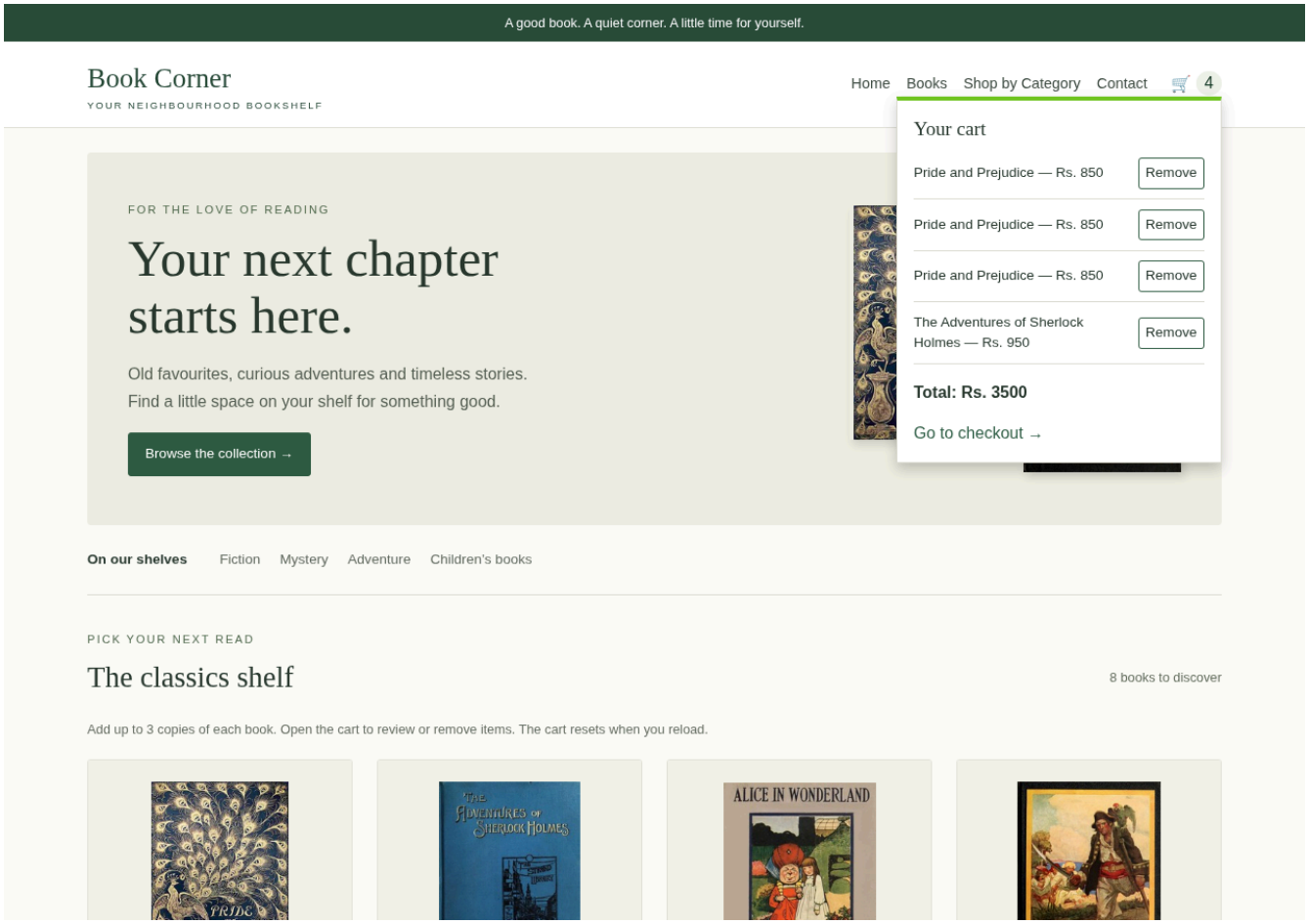
Task 3 — Cart after removal at 613px

One Pride and Prejudice copy removed: 3 items, total Rs. 2650. That book can be added again.



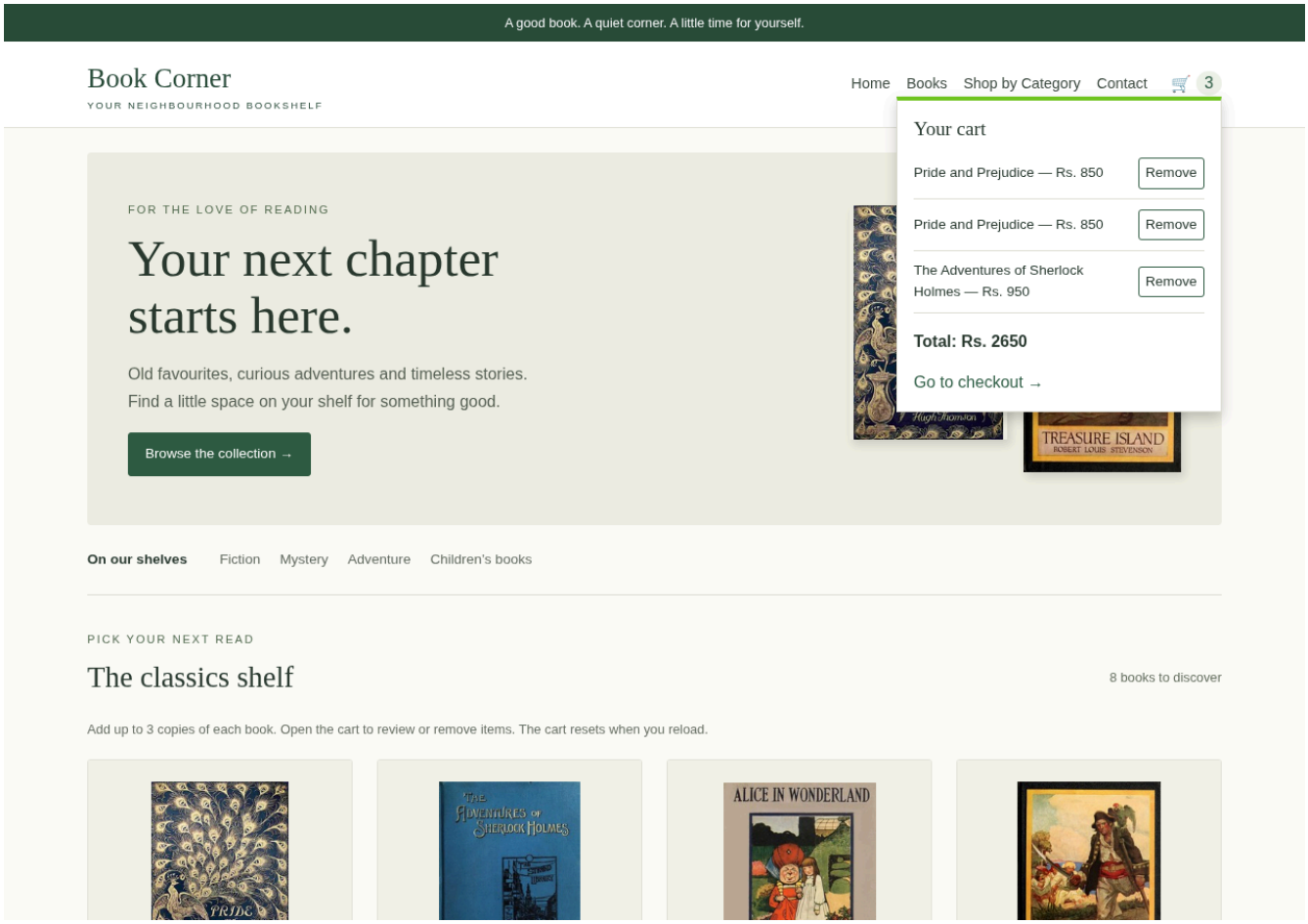
Task 3 — Cart after additions at 1280px

Three Pride and Prejudice copies and one Sherlock Holmes copy: 4 items, total Rs. 3500.



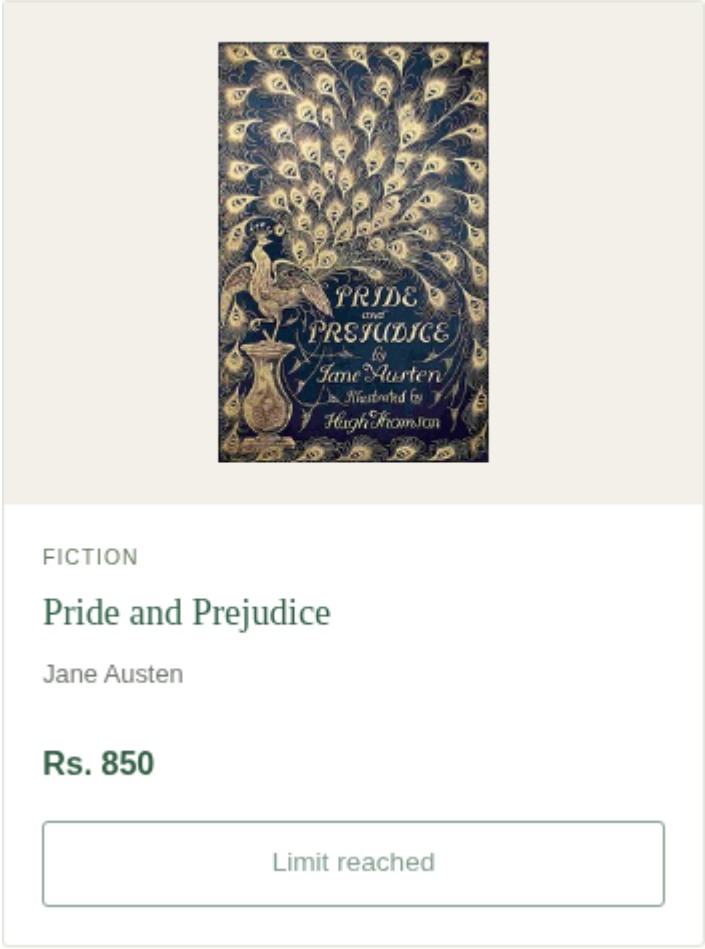
Task 3 — Cart after removal at 1280px

One Pride and Prejudice copy removed: 3 items, total Rs. 2650. That book can be added again.



Task 3 — Per-product limit

After three copies, the original card button reads Limit reached and is disabled.



Task 1 — Added or changed code

The shared header was updated in index.html and product.html at the Task 1 milestone. Task 3 later replaces the listing page cart link with the mini-cart markup shown in its own section.

index.html and product.html — header at Task 1

```
    <header class="site-header">
        <div class="mx-auto w-full max-w-[1140px] px-3 site-nav flex flex-wrap items-center justify-between">
            <a class="brand" href="index.html">Book Corner<span>YOUR NEIGHBOURHOOD BOOKSHELF</span></a>
            <input class="sr-only menu-toggle" type="checkbox" id="menu-toggle" aria-controls="main-menu">
            <label class="menu-button md:hidden rounded border border-[#deded5] px-3 py-2" for="menu-toggle"><span aria-
hidden="true">&#9776;</span><span class="sr-only">Toggle navigation</span></label>
            <nav class="menu-panel" id="main-menu" aria-label="Main navigation">
                <ul class="main-links flex w-full flex-col list-none p-0 m-0 md:flex-row md:w-auto">
                    <li class=""><a class="menu-link block px-2 py-2 no-underline" href="index.html">Home</a></li>
                    <li class=""><a class="menu-link block px-2 py-2 no-underline" href="index.html#books">Books</a></li>
                    <!-- Task 1: desktop hover/focus menu; mobile native accordions. -->
                    <li class="category-menu">
                        <a class="menu-link mega-trigger" href="index.html#categories">Shop by Category</a>
                        <div class="mega-menu" aria-label="Shop by Category">
                            <section>
                                <h2>Fiction</h2>
                                <ul>
                                    <li><a href="index.html#books">Classic fiction</a></li>
                                    <li><a href="index.html#books">Family stories</a></li>
                                    <li><a href="index.html#books">Romance</a></li>
                                    <li><a href="index.html#books">Literary fiction</a></li>
                                </ul>
                            </section>
                            <section>
                                <h2>Mystery</h2>
                                <ul>
                                    <li><a href="index.html#books">Detective stories</a></li>
                                    <li><a href="index.html#books">Sherlock Holmes</a></li>
                                    <li><a href="index.html#books">Gothic fiction</a></li>
                                    <li><a href="index.html#books">Suspense</a></li>
                                </ul>
                            </section>
                            <section>
                                <h2>Adventure</h2>
                                <ul>
                                    <li><a href="index.html#books">Sea adventures</a></li>
                                    <li><a href="index.html#books">Treasure hunts</a></li>
                                    <li><a href="index.html#books">Island stories</a></li>
                                    <li><a href="index.html#books">Voyages</a></li>
                                </ul>
                            </section>
                            <section>
                                <h2>Children’s books</h2>
                                <ul>
                                    <li><a href="index.html#books">Wonderland</a></li>
                                    <li><a href="index.html#books">Garden stories</a></li>
                                    <li><a href="index.html#books">Young readers</a></li>
                                    <li><a href="index.html#books">Fairy tales</a></li>
                                </ul>
                            </section>
                        </div>
                        <div class="category-accordion">
                            <details>
                                <summary>Fiction</summary>
                                <ul>
                                    <li><a href="index.html#books">Classic fiction</a></li>
                                    <li><a href="index.html#books">Family stories</a></li>
                                    <li><a href="index.html#books">Romance</a></li>
                                    <li><a href="index.html#books">Literary fiction</a></li>
                                </ul>
                            </details>
                            <details>
                                <summary>Mystery</summary>
                                <ul>
                                    <li><a href="index.html#books">Detective stories</a></li>
                                    <li><a href="index.html#books">Sherlock Holmes</a></li>
                                    <li><a href="index.html#books">Gothic fiction</a></li>
                                    <li><a href="index.html#books">Suspense</a></li>
                                </ul>
                            </details>
                            <details>
                                <summary>Adventure</summary>
```

```
                            <ul>
                                <li><a href="index.html#books">Sea adventures</a></li>
                                <li><a href="index.html#books">Treasure hunts</a></li>
                                <li><a href="index.html#books">Island stories</a></li>
                                <li><a href="index.html#books">Voyages</a></li>
                            </ul>
                        </details>
                        <details>
                            <summary>Children’s books</summary>
                            <ul>
                                <li><a href="index.html#books">Wonderland</a></li>
                                <li><a href="index.html#books">Garden stories</a></li>
                                <li><a href="index.html#books">Young readers</a></li>
                                <li><a href="index.html#books">Fairy tales</a></li>
                            </ul>
                        </details>
                    </div>
                </li>
                <li class=""><a class="menu-link block px-2 py-2 no-underline" href="#contact">Contact</a></li>
            </ul>
        </nav>
        <span class="cart" role="img" aria-label="Shopping cart, zero items">&#128722; <span>0</span></span>
    </div>
</header>
```

styles.css — Task 1 additions

```
/* Task 1. N = 13, BP = 613px, H = 91 degrees, K = 4 columns. */
:root { --accent: hsl(91, 70%, 45%); }
.site-header { position: relative; z-index: 1000; }
main { position: relative; z-index: 0; }
.site-nav { padding-top: 0; padding-bottom: 0; gap: 0 1rem; }
.brand { padding: 1rem 0; }
.menu-button { display: block; }
.menu-panel { display: none; margin-right: 0; }
.main-links { flex-direction: column; width: 100%; }
.mega-menu { display: none; }
.mega-trigger { display: block; padding: 0.6rem 0.5rem; text-decoration: none; }
.category-accordion { padding: 0 0.5rem 1rem; }
.category-accordion details { border-bottom: 1px solid #deded5; }
.category-accordion summary { padding: 0.6rem; cursor: pointer; }
.category-accordion ul, .mega-menu ul { list-style: none; padding: 0; margin: 0; }
.category-accordion li a, .mega-menu li a { display: block; padding: 0.35rem 0; font-size: 0.85rem; }
.category-accordion ul { padding: 0 1rem 0.6rem; }
a:focus-visible, summary:focus-visible, button:focus-visible { outline: 3px solid var(--accent); outline-offset: 2px; }

@media (min-width: 613px) {
    .menu-button, .menu-toggle, .category-accordion { display: none; }
    .menu-panel { display: flex; order: 4; flex-basis: 100%; align-self: stretch; }
    .main-links { flex-direction: row; width: auto; align-items: stretch; }
    .main-links > li { display: flex; align-items: stretch; }
    .main-links .menu-link { display: flex; align-items: center; padding: 1rem 0.5rem; }
    .mega-menu {
        position: absolute; top: 100%; left: 0; width: 100%; z-index: 30;
        grid-template-columns: repeat(4, minmax(0, 1fr)); gap: 1rem;
        padding: 1.25rem; background: white; border-top: 4px solid var(--accent);
        box-shadow: 0 6px 12px #0003;
    }
    .mega-menu h2 { font-size: 1rem; }
    .category-menu:hover .mega-menu,
    .category-menu:focus-within .mega-menu { display: grid; }
}

@media (min-width: 992px) {
    .menu-panel { order: 0; flex-basis: auto; }
}
```

Task 2 — Added or changed code

checkout.html — complete new page

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Checkout — Book Corner</title>
    <link rel="stylesheet" href="../tailwind.css">
    <link rel="stylesheet" href="styles.css">
</head>
<body>
    <div class="top-note">A good book. A quiet corner. A little time for yourself.</div>
    <header class="site-header">
        <div class="mx-auto w-full max-w-[1140px] px-3 site-nav flex flex-wrap items-center justify-between">
            <a class="brand" href="index.html">Book Corner<span>YOUR NEIGHBOURHOOD BOOKSHELF</span></a>
            <nav aria-label="Checkout navigation"><a href="index.html#books">Back to books</a></nav>
        </div>
    </header>
    <main class="mx-auto w-full max-w-[1140px] px-3 checkout-main">
        <h1>Checkout</h1>
        <p>All fields are required. This is a demo; no order will be placed.</p>
        <form class="checkout-form" action="checkout.html" method="get">
            <fieldset>
                <legend>Contact</legend>
                <div class="field-row">
                    <label for="full-name">Full name</label>
                    <div>
                        <input type="text" id="full-name" name="full-name" autocomplete="name" required aria-describedby="name-error">
                        <small class="field-error" id="name-error">Enter your full name.</small>
                    </div>
                </div>
                <div class="field-row">
                    <label for="email">Email</label>
                    <div>
                        <input type="email" id="email" name="email" autocomplete="email" required aria-describedby="email-error">
                        <small class="field-error" id="email-error">Enter a valid email address.</small>
                    </div>
                </div>
                <div class="field-row">
                    <label for="phone">Phone</label>
                    <div>
                        <input type="tel" id="phone" name="phone" autocomplete="tel" placeholder="0300-1234567" pattern="03[0-9]{2}-
[0-9]{7}" title="Use 03XX-XXXXXXX, for example 0300-1234567" required aria-describedby="phone-error">
                        <small class="field-error" id="phone-error">Use 03XX-XXXXXXX.</small>
                    </div>
                </div>
            </fieldset>
            <fieldset>
                <legend>Delivery</legend>
                <div class="field-row">
                    <label for="address">Address</label>
                    <div>
                        <input type="text" id="address" name="address" autocomplete="street-address" required aria-
describedby="address-error">
                        <small class="field-error" id="address-error">Enter your delivery address.</small>
                    </div>
                </div>
                <div class="field-row">
                    <label for="city">City</label>
                    <div>
                        <select id="city" name="city" autocomplete="address-level2" required aria-describedby="city-error">
                            <option value="">Choose a city</option>
                            <option>Lahore</option>
                            <option>Islamabad</option>
                            <option>Karachi</option>
                        </select>
                        <small class="field-error" id="city-error">Choose a delivery city.</small>
                    </div>
                </div>
                <div class="field-row">
                    <label for="delivery-date">Delivery date</label>
                    <div>
                        <!-- Exam date through exam date + N (13) days. -->
                        <input type="date" id="delivery-date" name="delivery-date" min="2026-10-29" max="2026-11-11" required aria-
describedby="date-hint date-error">
                        <small id="date-hint">29 Oct – 11 Nov 2026</small>
```

```
                        <small class="field-error" id="date-error">Choose 29 Oct – 11 Nov 2026.</small>
                    </div>
                </div>
            </fieldset>
            <button class="button checkout-submit" type="submit">Submit checkout</button>
        </form>
    </main>
    <footer id="contact">
        <div class="mx-auto w-full max-w-[1140px] px-3">
            <div class="grid grid-cols-12 gap-6">
                <div class="col-span-12 md:col-span-6"><h2 class="footer-brand">Book Corner</h2><p>A small collection of stories worth
making time for.</p><p>A student storefront built for Web Technologies.</p></div>
                <div class="col-span-6 md:col-span-3"><h2>Explore</h2><ul><li><a href="index.html">Home</a></li><li><a
href="index.html#books">All books</a></li><li><a href="index.html#categories">Categories</a></li></ul></div>
                <div class="col-span-6 md:col-span-3"><h2>Get in touch</h2><ul><li><a href="mailto:yarmuhammadawaim@gmail.com">Email
us</a></li><li><a href="../assignment-1/index.html">About the student</a></li></ul><p class="text-sm">Static coursework demo. Orders
are not accepted yet.</p></div>
            </div>
            <p class="copyright">&copy; 2026 Book Corner · CSC336 Midterm</p>
        </div>
    </footer>
</body>
</html>
```

index.html and product.html — checkout link added at Task 2

```
<a class="cart" href="checkout.html" aria-label="Go to checkout">&#128722; <span>0</span></a>
```

styles.css — Task 2 additions

```
/* Task 2: HTML validates the form; CSS only controls layout and feedback. */
.checkout-main { padding-top: 2rem; padding-bottom: 3rem; }
.checkout-main h1 { font-size: 2rem; }
.checkout-form { display: grid; gap: 1.25rem; }
.checkout-form fieldset { min-width: 0; padding: 1rem; border: 1px solid #deded5; background: white; }
.checkout-form legend { padding: 0 0.4rem; font-size: 1.2rem; font-weight: bold; }
.field-row { display: grid; gap: 0.4rem; margin-bottom: 1rem; }
.field-row label { font-size: 0.9rem; padding-top: 0.5rem; }
.field-row input, .field-row select {
    width: 100%; min-width: 0; padding: 0.5rem; border: 1px solid #9baba0;
    border-radius: 3px; background: white; font-size: 0.85rem;
}
.field-row input:focus, .field-row select:focus { outline: 2px solid var(--accent); }
.field-row small { font-size: 0.75rem; }
.field-error { display: none; color: #b42318; }
.field-row input:user-invalid, .field-row select:user-invalid { border-color: #b42318; }
.field-row input:user-invalid ~ .field-error,
.field-row select:user-invalid ~ .field-error { display: block; }
.checkout-submit {
    grid-column: 1 / -1; width: 100%; background: var(--accent);
    color: #183c29; border: 1px solid #315c43; cursor: pointer;
}
@media (min-width: 613px) {
    .checkout-form { grid-template-columns: repeat(2, minmax(0, 1fr)); }
    .field-row { grid-template-columns: 5rem minmax(0, 1fr); gap: 0.6rem; }
}
```

Task 3 — Added or changed code

index.html — script loading

```
<script src="script.js" defer></script>
```

index.html — mini cart replacing the old cart link

```
            <div class="cart-area">
                <a class="cart" id="cart-toggle" href="checkout.html" aria-label="Shopping cart" aria-controls="mini-cart" aria-
expanded="false">&#128722; <span id="cart-count" aria-live="polite">0</span></a>
                <section class="mini-cart" id="mini-cart" aria-labelledby="mini-cart-title" hidden>
                    <h2 id="mini-cart-title">Your cart</h2>
                    <ul id="cart-items"><li>Your cart is empty.</li></ul>
                    <p class="cart-total">Total: Rs. <span id="cart-total">0</span></p>
                    <a href="checkout.html">Go to checkout &rarr;</a>
                </section>
            </div>
```

index.html — updated catalogue note

```
<p class="demo-note" id="cart-note">Add up to 3 copies of each book. Open the cart to review or remove items. The cart resets when you
reload.</p>
```

script.js — complete new file

```
// N = 13, so each product can be added (13 % 4) + 2 = 3 times.
const limit = (13 % 4) + 2;
const cart = [];
const cards = document.querySelectorAll('.book-card');
const cartToggle = document.querySelector('#cart-toggle');
const miniCart = document.querySelector('#mini-cart');
const cartItems = document.querySelector('#cart-items');
const cartCount = document.querySelector('#cart-count');
const cartTotal = document.querySelector('#cart-total');

// Rebuild the small list and keep the badge, total and buttons in sync.
function updateCart() {
    cartCount.textContent = cart.length;
    cartTotal.textContent = cart.reduce((total, item) => total + item.price, 0);
    cartItems.textContent = '';

    if (cart.length === 0) {
        const emptyMessage = document.createElement('li');
        emptyMessage.textContent = 'Your cart is empty.';
        cartItems.append(emptyMessage);
    }

    cart.forEach((item, index) => {
        const row = document.createElement('li');
        const description = document.createElement('span');
        description.textContent = `${item.name} — Rs. ${item.price}`;
        const removeButton = document.createElement('button');
        removeButton.type = 'button';
        removeButton.textContent = 'Remove';
        removeButton.setAttribute('aria-label', `Remove ${item.name}`);
        removeButton.addEventListener('click', () => {
            cart.splice(index, 1);
            updateCart();
        });
        row.append(description, removeButton);
        cartItems.append(row);
    });

    cards.forEach(card => {
        const name = card.querySelector('.book-title').textContent.trim();
        const count = cart.filter(item => item.name === name).length;
        const button = card.querySelector('button');
        button.disabled = count >= limit;
        button.textContent = button.disabled ? 'Limit reached' : 'Add to Cart';
    });
}
```

```
cards.forEach(card => {
    const button = card.querySelector('button');
    button.addEventListener('click', () => {
        const name = card.querySelector('.book-title').textContent.trim();
        const price = Number(card.querySelector('.price').textContent.replace('Rs.', '').trim());
        const count = cart.filter(item => item.name === name).length;
        if (count >= limit) return;
        cart.push({ name, price });
        updateCart();
    });
});

cartToggle.addEventListener('click', event => {
    event.preventDefault();
    miniCart.hidden = !miniCart.hidden;
    cartToggle.setAttribute('aria-expanded', String(!miniCart.hidden));
});

updateCart();
```

styles.css — Task 3 additions

```
/* Task 3: a small panel anchored under the cart icon. */
.cart-area { position: relative; align-self: center; }
.mini-cart {
    position: absolute; top: 100%; right: 0; z-index: 20;
    width: min(320px, calc(100vw - 24px)); padding: 1rem;
    background: white; border: 1px solid #deded5; border-top: 4px solid var(--accent);
    box-shadow: 0 6px 12px #0003;
}
.mini-cart h2 { font-size: 1.2rem; }
.mini-cart ul { list-style: none; margin: 0 0 1rem; padding: 0; max-height: 220px; overflow-y: auto; }
.mini-cart li { display: flex; justify-content: space-between; align-items: center; gap: 0.75rem; padding: 0.6rem 0; border-bottom:
1px solid #deded5; font-size: 0.85rem; }
.mini-cart li span { min-width: 0; overflow-wrap: anywhere; }
.mini-cart button { flex-shrink: 0; border: 1px solid #315c43; border-radius: 3px; padding: 0.25rem 0.4rem; cursor: pointer; }
.cart-total { font-weight: bold; }
.book-card button:disabled { opacity: 0.6; cursor: not-allowed; }
```